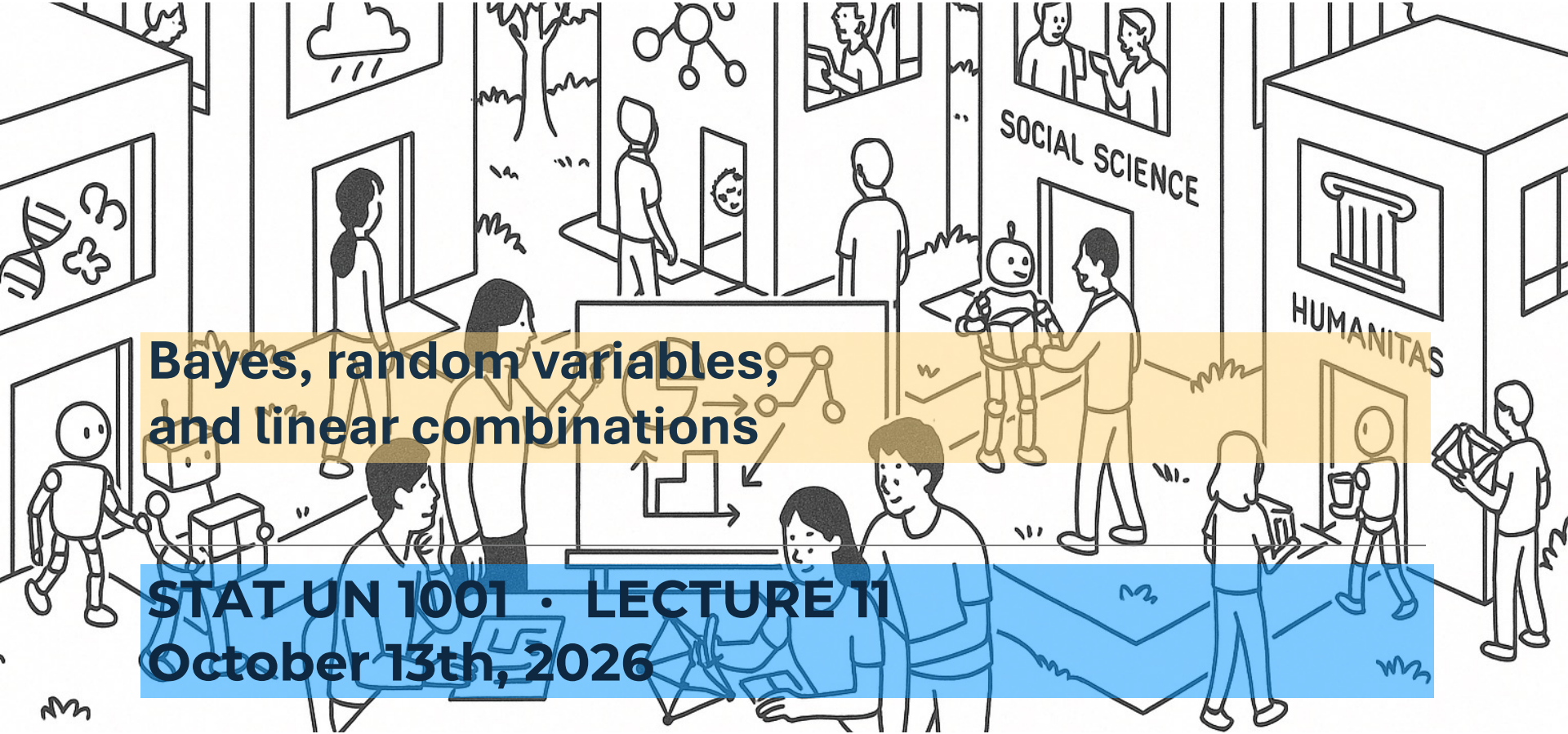


Bayes, random variables, and linear combinations

STAT UN 1001 · LECTURE 11
October 13th, 2026

From flipped activities check-in

REQUESTS FOR REVIEW

Topic	Student signal
Bayes' rule	33 check-in requests
P(A | B) from the tree	55% correct
Linear combinations	shift plus scale needs another pass

TODAY'S AIM

Use the story and the denominator before using formulas.

The five-part setup

1	**Chance process**	What random action repeats?
2	**Sample space**	What outcomes could happen?
3	**Event**	Which outcomes count?
4	**Probability sentence**	What probability are we finding?
5	**Rule**	Tree, table, complement, or multiplication?

Exactly two successes in three tries

Chance of success each time is 0.5. We want exactly two successes in three tries.

Path	Probability
SSW	0.5 × 0.5 × 0.5 = 0.125
SWS	0.5 × 0.5 × 0.5 = 0.125
WSS	0.5 × 0.5 × 0.5 = 0.125

3 paths × 0.125 = 0.375

Practice quiz tree

A = household has a cat B = household bought pet food

First branch	Probability	Second branch	Probability
A	P(A) = 0.30	B given A	P(B | A) = 0.80
not A	P(not A) = 0.70	B given not A	P(B | not A) = 0.20

Question: among households that bought pet food, what fraction have a cat?

Rebuilding the denominator

We restrict attention to households that bought pet food.

Path	Probability
cat and bought food	0.30 × 0.80 = 0.24
no cat and bought food	0.70 × 0.20 = 0.14
all bought food	0.24 + 0.14 = 0.38

$$P(A \mid B) = 0.24 / 0.38 = 0.63$$

P(B | A) and P(A | B)

Expression	Words	Denominator
P(B | A)	Among households with cats, how many bought pet food?	households with cats
P(A | B)	Among households that bought pet food, how many have cats?	households that bought pet food

Same events, different denominator, different question.

Guided Practice: Phone update and typos

SETUP

New software: P(typo)=30%.
Old software: P(typo)=10%.
Your friend has typos in two recent texts.

WHAT WE NEED BEFORE COMPUTING

Prior chance that she updated her phone.

ASSUMPTIONS TO DISCUSS

Do the reported typo rates apply to this friend?
The two texts behave independently given the phone system.
The definition of typo stays consistent.

Phone update tree

U = updated phone T = two recent texts have typos

Branch	Probability
U	P(U) = p
not U	P(not U) = 1 − p

Typo evidence	Probability
T given U	0.30 × 0.30 = 0.09
T given not U	0.10 × 0.10 = 0.01

P(U | T) depends on both the evidence and the prior p.

The prior changes the answer

Prior P(U)	P(U and T)	P(T)	P(U | T)
0.10	0.10×0.09 = 0.009	0.009 + 0.90×0.01 = 0.018	0.50
0.20	0.20×0.09 = 0.018	0.018 + 0.80×0.01 = 0.026	0.69
0.50	0.50×0.09 = 0.045	0.045 + 0.50×0.01 = 0.050	0.90

The same two typo texts lead to different posteriors when the prior changes.

Bayes' Rule Saves Lives – Sally Clark SIDS Case



- In November 1999, Sally Clark, an English solicitor, was convicted of murdering her infant sons. The first, Christopher, had been 11 weeks old when he died, in 1996, and the second, Harry, 8 weeks old, in January 1998, when he was found dead. Christopher was believed to have been a victim of "cot death", called SIDS (Sudden Infant Death Syndrome) in the US. After her second baby, Harry, also died in his crib, Sally Clark was arrested for murder.
- The star witness for the prosecution was a well known pediatrician and professor, Sir Roy Meadow, who authored the infamous Meadow's Law :"One sudden infant death is a tragedy, two is suspicious and three is murder until proved otherwise"[2].
- The Royal Statistical Society protested the misuse of statistics in courts, but not early enough to prevent Sally Clark's conviction. She was eventually acquitted and released, only to die at the age of 42 through alcohol poisoning.

https://stat20.berkeley.edu/spring-2024/3-probability/02-cond-prob-indep/notes.html

Bayes' Rule Saves Lives – Sally Clark SIDS Case



- The math presented by Meadow, in brief: Based on various studies, there is a probability of 1 in 8,543 of a baby dying of SIDS in a family such as the Clarks. As the Clarks suffered *two* deaths, Meadow multiplied 8,543 by 8,543 to arrive at 73 million. He told the jury that the chance or probability that the event of two "cot deaths" was 1 in 73 million.
- The first error was in treating the deaths as independent. It turns out that the probability of a second child dying of "cot death" or SIDS is 1/60 *given* that the first child died of SIDS.
- The second error is an example of what is called the Prosecutor's Fallacy. What is needed is P(innocence | evidence), but it is often confused with P(evidence | innocence).
- The prosecutor failed to recognize the low chance of murders.

https://stat20.berkeley.edu/spring-2024/3-probability/02-cond-prob-indep/notes.html

Random variables

A random variable assigns a number to each outcome of a chance process.

Example: X = number of typo-containing messages among two texts.

Outcome	X
no typo, no typo	0
typo, no typo	1
no typo, typo	1
typo, typo	2

Expected typo-containing texts

System	P(X=0)	P(X=1)	P(X=2)	E(X)
Old	0.81	0.18	0.01	0.20
New	0.49	0.42	0.09	0.60

E(X) = sum of value × probability

Interpretation: over many pairs of texts, the new system averages 0.60 typo-containing texts per pair.

Expected value interpretation

ONE RUN

One pair of texts may have 0, 1, or 2 typo-containing messages.

MANY RUNS

The expected value describes the long-run average under the same chance process.

Practice: if X has values 0, 1, 2 with probabilities .25, .50, .25, what is E(X)?

Linear combinations

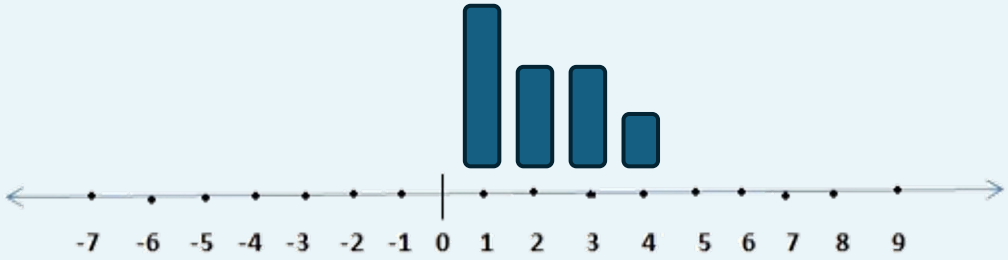
If $Y = aX + b$, then

Quantity	Rule	Meaning
Mean	$E(Y) = aE(X) + b$	shift and scale affect center
Standard deviation	$SD(Y) = \|a\|SD(X)$	scale affects spread
Adding b	no change to SD	a shift moves values together

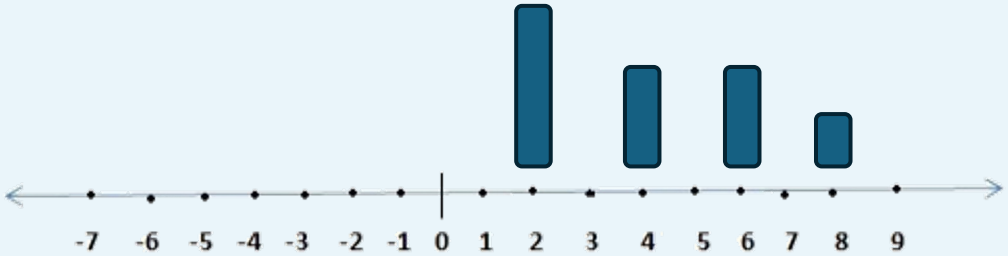
This is OpenIntro Section 3.4: linear combinations of random variables.

Linear combinations

Random Variable X



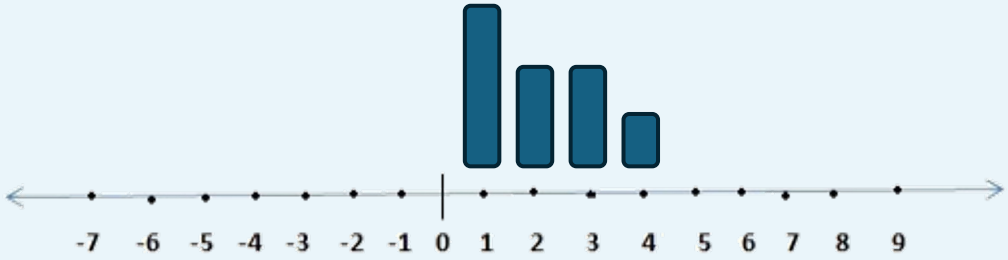
Random Variable Y = 2X



Mean of Y = 2 (mean of X)
std. dev of Y = 2 (std. dev of X)

Linear combinations

Random Variable X



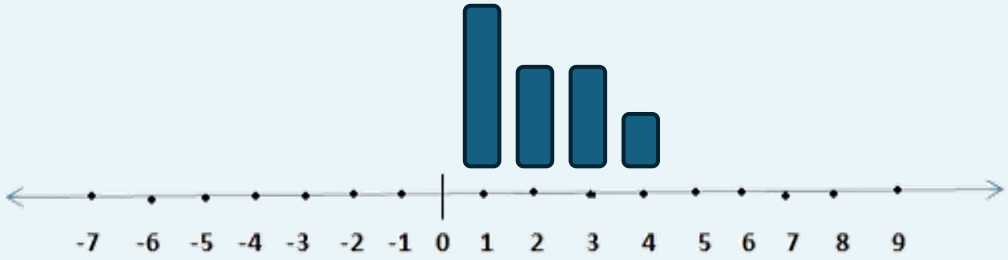
Random Variable Y = X+1



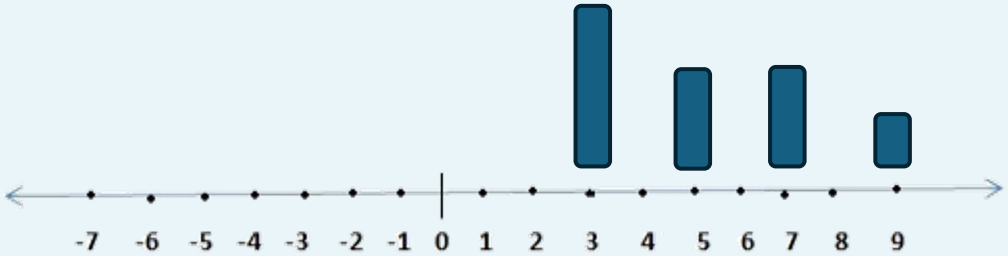
Mean of Y = (mean of X) + 1
std. dev of Y = (std. dev of X)

Linear combinations

Random Variable X



Random Variable Y = 2X+1



Mean of Y = 2 (mean of X) + 1
std. dev of Y = 2 (std. dev of X)

Celsius to Fahrenheit

Daily temperature C has mean 20 and SD 3. $F = 1.8C + 32$.

Quantity	Computation	Answer
Mean of F	$1.8 \times 20 + 32$	68
SD of F	1.8×3	5.4

Common mistake

Adding 32 changes every value by the same amount, so it does not change spread.